

AYMANE AIT DADS

Research Intern - Medical Imaging | Image Classification · Supervised Learning · Dataset Curation
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PROFESSIONAL SUMMARY

Ranked 1st in EURECOM class (0.791 AUC) on an AI-generated image detection benchmark submitted to NTIRE 2026 @ CVPR, demonstrating direct capability in classification and detection tasks central to medical imaging AI. Automated analysis of 100K+ fiber-optic network records at Orange Maroc using ensemble ML pipelines, reducing manual analysis time by ~60% and delivering structured outputs directly used by operational teams. Brings hands-on experience in supervised learning, multimodal image classification, dataset curation, quality control, and model evaluation - aligning precisely with the collaboration between AI scientists and clinical researchers at Siemens Healthineers. Aims to contribute to diagnostic-quality AI validation and structured reporting pipelines for detection, segmentation, and quantification tasks across chest X-ray, CT, and MRI modalities.

CORE COMPETENCIES

supervised learning | image classification | detection & segmentation | dataset curation | quality control | model performance validation | multimodal imaging data | structured reporting

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built Random Forest and K-Means classification models on **100K+ fiber-optic network records**, categorizing node quality across upload speed, latency, and jitter dimensions.
- Developed a clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team for maintenance decisions.
- Reduced manual analysis time by **~60%** through automated feature engineering, model evaluation workflows, and reproducible data pipelines built in Python and Scikit-learn.
- Delivered structured stakeholder presentations in Power BI translating model outputs into **actionable maintenance recommendations**, bridging technical findings with operational decision-making.

PROJECTS

AI-Generated Image Detection

EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in Class

- Ranked **1st in EURECOM class** private Kaggle leaderboard (0.791 AUC) for AI-generated image detection; independently submitted to the NTIRE 2026 @ CVPR CodaBench international benchmark for external validation.
- Fine-tuned CLIP ViT-L/14 (428M parameters) with a custom MLP classification head on **250K images spanning 25+ generator types**, applying 10-view TTA and AUC-weighted ensembling to maximize out-of-distribution generalization.

PyTorch · OpenCLIP · ViT · FP16 Mixed Precision · Kaggle GPU

Aerial Cactus Detection - Endangered Species Classification

EURECOM Computer Vision · 2025 · 99.8% Accuracy

- Benchmarked CNN, DenseNet121, and hybrid CNN-Transformer architectures on **17,500+ drone images** for endangered species classification, achieving 99.8% accuracy, F1 = 0.999, and ROC AUC = 1.0.
- Designed a rigorous model comparison protocol with structured quality control across all three architectures, producing **validated performance metrics** directly comparable to supervised learning evaluation standards in medical imaging.

PyTorch · DenseNet121 · CNN · Scikit-learn · Matplotlib

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Image Security, NLP, Statistics, Cloud Computing

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

Computer Vision & Imaging: image classification | detection & segmentation | CLIP ViT | CNN | DenseNet | OpenCLIP | TTA ensemble

ML & Data Science: PyTorch | Scikit-learn | Pandas | feature engineering | supervised learning | ensemble methods | quality control

Tools & Reporting: Python | Power BI | Git | Matplotlib | Seaborn | SQL | Linux